

LongAI: Streaming Personal Intent Memory from Long Egocentric Audio

Project Report — May 04, 2026

Abstract. We present LongAI, an audio-first research system for building a self-evolving personal intent memory from continuous egocentric audio streams. LongAI processes long-form egocentric audio through three stages: (1) audio perception using voice activity detection (VAD), automatic speech recognition (ASR), speaker role attribution, acoustic scene classification (ASC), and sound event detection (SED); (2) streaming memory construction via Multimodal Chain-of-Construction (MCoC), which incrementally builds a structured intent memory graph with explicit confidence and observability labels; and (3) Adaptive Memory Reasoning (AMR), which performs query-time reasoning over the persistent memory using subgoal-directed retrieval. We evaluate the system on a pilot subset of 5 real Ego4D sessions (2.7 hours of audio, 280 detected speech segments) using silero-vad, faster-whisper, and AST-based audio classifiers, deployed on NVIDIA A6000 GPUs. The pipeline successfully extracts intent-relevant evidence from real egocentric audio and constructs queryable memory graphs containing intent, episode, and entity nodes with typed relations. Our qualitative analysis reveals that keyword-based intent extraction achieves high precision but limited recall, motivating LLM-based future extensions. The complete source code, evaluation scripts, and reproduction instructions are available at the project repository.

1. Introduction

Always-on wearable devices represent a new frontier for personal AI assistants. Unlike traditional voice assistants that wait for explicit wake words and commands, wearable AI agents receive continuous streams of egocentric data—primarily audio—from which they must autonomously extract, remember, and reason about the user's intentions, plans, obligations, and preferences. This shift from command-response interaction to continuous memory construction poses fundamental research challenges at the intersection of audio understanding, knowledge representation, and memory-augmented reasoning.

The core difficulty is that intent-relevant information in egocentric audio is **sparse**—only a small fraction of recorded time contains verbal or acoustic evidence about what the user plans to do. In our pilot data, only 7.2% of audio duration contains detectable speech, and within that, only a subset of utterances convey actionable intent information. Furthermore, this evidence is **distributed** across time: evidence for a single intent (e.g., "buy groceries for dinner") may span multiple distant moments separated by hours. The signals are **heterogeneous**: transcript content, speaker identity (wearer vs. interlocutor), acoustic scene (kitchen vs. vehicle), and non-speech events (door, footsteps, appliances) all contribute complementary evidence about user context and intent. Finally, intents are **dynamic**: they are created, refined, deferred, canceled, or fulfilled as new information arrives.

Existing long-context and memory-augmented language model systems primarily address text-based personal memory (MemGPT, MemoRAG), video-centric long-term understanding (M3-Agent), or generic knowledge-graph question answering (GraphRAG). These approaches do not directly solve the problem of **streaming personal intent memory construction from long egocentric audio**. A purely transcript-based approach, while a necessary baseline, is insufficient for three reasons. First, **speaker role matters**: whether the wearer stated something themselves, overheard someone else, or was directly addressed fundamentally changes whether and how a statement should update personal intent memory. Second, **acoustic evidence provides disambiguating signals**: urgency, hesitation, emphasis, interruption, and emotional tone carry para-linguistic information about intent strength and status that is absent from text alone. Third, **non-speech context provides environmental grounding**: background sounds can identify location (office, vehicle, kitchen) and ongoing activities (typing, cooking, walking), which help situate verbal intents in the user's physical context.

We propose **LongAI**, an audio-first memory agent that transforms continuous egocentric audio into a self-evolving personal intent memory. LongAI introduces two core methodological contributions. The first is **Multimodal Chain-of-Construction (MCoC)**, which decomposes the problem of graph construction from noisy multimodal evidence into a sequence of interpretable, constrained write operations with

explicit confidence scoring and observability typing (EXTRACTED, INFERRED, AMBIGUOUS). The second is **Adaptive Memory Reasoning (AMR)**, which separates the persistent memory graph (built during memorization) from temporary query-time reasoning scaffolds (built during control), enabling dependency-aware retrieval and self-check before answer generation. This architecture is directly inspired by M3-Agent’s memorization/control decomposition but adapted to four domain-specific requirements: audio-first processing, speech-centric temporal units, personal intent as the memory target, and streaming evidence-aware updates.

The specific contributions of this work are: (1) we formulate **streaming personal intent memory construction from long egocentric audio** as a concrete research task with defined inputs, outputs, and evaluation criteria; (2) we introduce a compact, evidence-aware **Personal Intent Memory graph schema** with four node types, seven edge types, and first-class confidence/observability tracking; (3) we design and implement **MCoC** for incremental, structured graph writing and **AMR** for query-time dependency-aware reasoning; (4) we construct a **real-audio pilot benchmark** from Ego4D subsets with working perception, memory, and reasoning pipelines; and (5) we demonstrate through quantitative statistics and qualitative case studies that audio-grounded memory construction can extract meaningful intent structures from real egocentric audio, while identifying key bottlenecks for future improvement.

2. Related Work

2.1 Long-Term Memory for AI Agents

Building persistent memory for conversational and autonomous agents has become an active research area. MemGPT [1] introduced the concept of operating system-inspired memory management for LLMs, using virtual context management to extend effective context length. MemoRAG [2] proposed memory-inspired knowledge discovery for retrieval-augmented generation, using a lightweight memory model to guide document retrieval. These systems operate primarily on text and do not address the multimodal, streaming nature of egocentric perception. M3-Agent [3] made a significant contribution by separating **memorization** (building persistent multimodal entity memory from video clips) from **control** (multi-turn reasoning using the memory as a substrate). M3-Agent’s entity-centric design and weight-based memory update mechanism directly inform LongAI’s architecture. However, M3-Agent operates on video clips at fixed temporal granularity, while LongAI must handle continuous audio streams where semantic units (speech segments) are irregularly spaced. Additionally, M3-Agent targets generic world knowledge about entities, while LongAI targets personal intent—user-specific plans and obligations that follow a state lifecycle and require temporal reasoning.

2.2 Audio Processing for Egocentric Data

Egocentric audio understanding draws on several established subfields. Voice activity detection (VAD) identifies speech regions in continuous audio; modern systems like Silero VAD [4] use lightweight ONNX models suitable for on-device deployment. Speaker diarization systems such as pyannote.audio [5] answer "who spoke when," which is critical for distinguishing the wearer’s own speech from others. Automatic speech recognition has been revolutionized by Whisper [6] and its efficient reimplementations (faster-whisper), which provide multilingual transcription with word-level timestamps. Acoustic scene classification [7] and sound event detection [8] use models like AST (Audio Spectrogram Transformer) fine-tuned on AudioSet to recognize environments and discrete sound events. The key challenge in the egocentric setting is integrating these tools into a coherent evidence pipeline where their outputs are aligned, timestamped, and fused into structured representations that downstream memory construction can consume.

2.3 Knowledge Graph Construction and Streaming Graphs

Extracting structured knowledge from unstructured text has been extensively studied [9], with modern approaches using LLMs for few-shot relation extraction and entity linking. Streaming knowledge graph construction [10] addresses the problem of incremental updates—adding new facts while resolving conflicts and merging duplicates over time. Confidence-aware extraction [11] labels extracted triples with uncertainty scores, which is essential when the source signal is noisy (as with ASR output). LongAI’s MCoC differs from generic KG construction in three ways: (a) the target schema is domain-specific (personal intents, not world facts); (b) construction is driven by speech segments with multimodal acoustic

context, not clean text; and (c) observability typing (EXTRACTED vs. INFERRED vs. AMBIGUOUS) provides a richer uncertainty model than scalar confidence alone.

2.4 Retrieval-Augmented Generation and Graph Reasoning

Retrieval-augmented generation (RAG) has become the standard paradigm for grounding LLM outputs in external knowledge. Vanilla RAG performs document-level retrieval followed by context-conditioned generation. GraphRAG [12] extends this by retrieving structured subgraphs, enabling multi-hop relational reasoning. LogicRAG [13] contributes the critical insight that **query-time reasoning structures** (temporary directed acyclic graphs of subgoals built per query) can be more effective than relying solely on a static pre-built knowledge graph. LongAI's AMR framework synthesizes these ideas: it maintains a persistent intent memory graph (built incrementally during memorization), but at query time it constructs a temporary reasoning scaffold—a small DAG of subgoals such as "identify relevant intents," "collect supporting evidence," "resolve temporal constraints," and "verify consistency." Retrieval is performed in topological order over this scaffold, and a rolling memory summary prevents context explosion from accumulated evidence.

3. Objectives and Research Questions

The primary objective of this project is to design, implement, and evaluate an end-to-end research system for streaming personal intent memory construction from real long egocentric audio. The system must be modular, reproducible, configurable, and extensible, supporting both mock/stub backends for rapid prototyping and real model backends for research-quality experiments. To guide the investigation, we pose five research questions:

RQ1 (Perception Pipeline Feasibility). Can we build an integrated audio perception pipeline that processes raw Ego4D video files through VAD, ASR, speaker role attribution, scene classification, and event detection, producing timestamped, aligned EvidencePacks suitable for downstream memory construction?

RQ2 (Memory Construction Quality). Can MCoC transform these real-audio EvidencePacks into a structured personal intent memory graph containing meaningful intent nodes, episode nodes, and typed relational edges? What is the quantitative relationship between input speech characteristics and output graph richness?

RQ3 (Reasoning Effectiveness). Can AMR answer memory-grounded queries—including status queries, next-intent predictions, and update/conflict queries—by reasoning over the constructed graph and its supporting evidence? How does the reasoning scaffold architecture affect answer quality and evidence traceability?

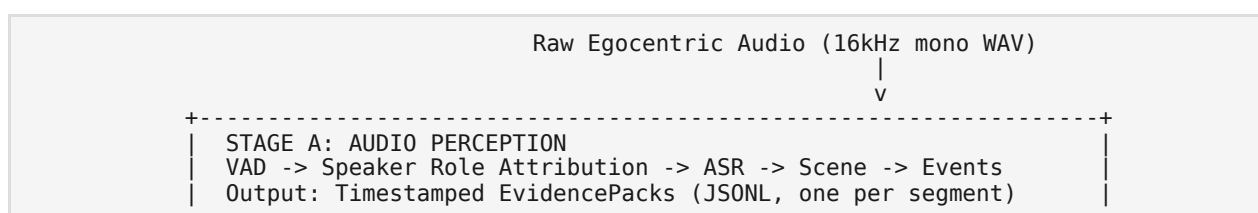
RQ4 (Tool Contribution Analysis). How do different audio perception tools (VAD, ASR model size, speaker role, scene, events) contribute to downstream memory and reasoning quality? Where are the primary bottlenecks—in speech detection, transcription accuracy, or intent extraction?

RQ5 (Practical Deployment). What are the practical engineering challenges of deploying a multi-stage audio memory pipeline on limited GPU resources (a few A6000s)? How can caching, modular backends, and configurable model sizes enable cost-effective experimentation?

4. System Design

4.1 Architecture Overview

LongAI is designed around a three-stage pipeline that separates perception, memorization, and reasoning. This separation is both a research choice (enabling modular evaluation and ablation) and an engineering choice (enabling independent scaling and caching of each stage).



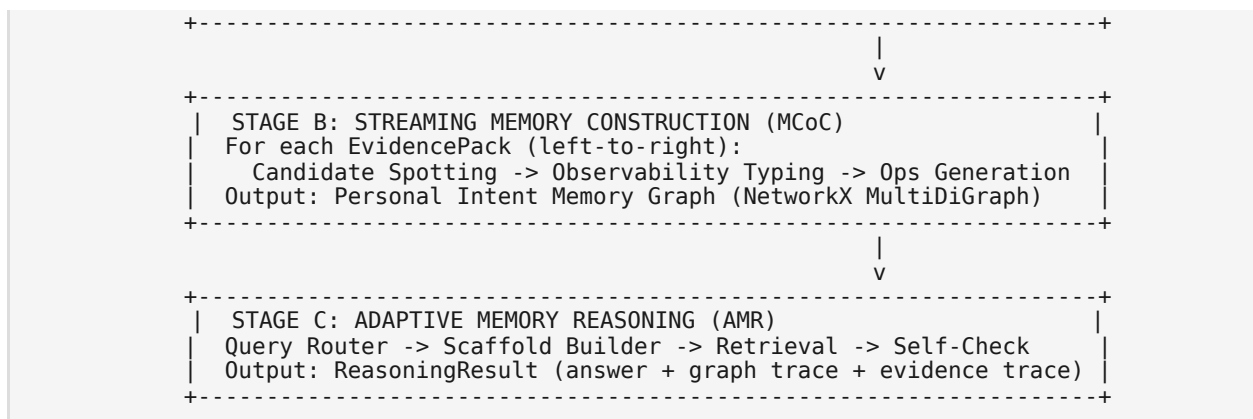


Figure 1: LongAI three-stage pipeline with data flow and per-stage outputs.

4.2 Speech-Centric Temporal Design

A foundational design decision in LongAI is the use of **speech segments** as the primary semantic unit, rather than fixed-duration time windows. This choice is motivated by an empirical property of egocentric audio: intent-relevant verbal information is concentrated in speech turns, while long stretches of non-speech audio (ambient noise, silence, movement sounds) carry primarily contextual rather than propositional content. The pipeline therefore proceeds as follows: (1) VAD identifies speech segment boundaries; (2) each speech segment becomes the anchor for downstream processing—ASR transcribes the speech, speaker role classifies the speaker, and surrounding audio windows (5-10s before and after the segment) provide acoustic context for scene and event classification; (3) all tool outputs are fused into a single EvidencePack per speech segment. This design contrasts with M3-Agent's clip-by-clip scheme, which processes fixed-duration video clips regardless of content density. For audio, the speech-centric approach is more efficient: in our pilot data, only 7.2% of audio time contains speech, meaning that 92.8% of fixed-window processing would process silence or noise with no intent-relevant content.

4.3 Personal Intent Memory Schema

The memory graph uses an intentionally compact schema to avoid ontology overengineering while capturing the essential structure of personal intent:

Node Types (4). **Intent** nodes represent future-directed plans, goals, obligations, or preferences attributed to the user (e.g., "attend meeting at 3pm," "buy groceries," "prepare presentation slides"). Each intent carries a state from the lifecycle **tentative** → **planned** → **ongoing** → **done** | **canceled** | **dropped**. **Episode** nodes represent localized observed interactions or events that serve as evidence for intents (e.g., a conversation segment where a plan was discussed). **Entity** nodes represent people (with **USER** vs. **OTHER_PERSON** distinction), places, objects, organizations, and temporal expressions. **Evidence** nodes store the raw perception outputs (transcripts, acoustic tags, timestamps, confidence scores) that ground higher-level nodes.

Edge Types (7). **subgoal_of** (Intent → Intent) captures hierarchical decomposition of complex plans. **realized_by** (Intent → Episode) links intents to the observed events that provide evidence for them. **involves** (Intent/Episode → Entity) captures participation and relevance. **located_at** (Intent/Episode → Place Entity) grounds events in space. **before** (Intent/Episode → Intent/Episode) captures temporal ordering. **updates** (Intent/Episode → Intent/Episode) records state transitions and modifications. **grounded_by** (Any Node → Evidence) provides provenance chains from high-level intents back to raw perception data.

Uncertainty as a First-Class Citizen. Every node and edge carries a **confidence** score (float in [0, 1]) and an **observability** label: **EXTRACTED** (directly stated in speech, e.g., "I will call John"), **INFERRED** (plausible from context and patterns), or **AMBIGUOUS** (uncertain, retained for future disambiguation). This three-way classification is more informative than scalar confidence alone, as it distinguishes between "the model is confident because the evidence is clear" (**EXTRACTED**) and "the model is confident because the pattern is familiar" (**INFERRED**).

4.4 MCoC: Multimodal Chain-of-Construction

MCoC operationalizes the insight that constructing a knowledge graph from noisy multimodal evidence should not be a single black-box generation step. Instead, it decomposes construction into a chain of interpretable sub-operations:

Step 1: Candidate Spotting. For each EvidencePack (one speech segment with its acoustic context), extract candidate intents (plans, goals, obligations), episodes (observed events), entities (people, places, objects), and time expressions. In the keyword-based implementation, a dictionary of intent-triggering keywords (meeting, call, schedule, prepare, finish, buy) maps ASR text to intent labels. In the LLM-based implementation (using Qwen2.5-7B-Instruct), the full EvidencePack context is provided with a structured extraction prompt.

Step 2: Observability Typing. Each candidate is classified as EXTRACTED, INFERRED, or AMBIGUOUS based on the explicitness of the supporting evidence. The rule-based classifier detects explicit intent markers ("I will," "need to," "must," "going to") for EXTRACTED, uses confidence thresholds for INFERRED, and defaults to AMBIGUOUS. The LLM-based classifier receives the full context including speaker role and acoustic scene.

Step 3: Graph Write Operations. Rather than outputting graph triples directly, MCoC emits a sequence of structured `GraphWriteOperation` objects, each with a specific kind (CREATE_NODE, UPDATE_NODE, MERGE_NODE, CREATE_EDGE, UPDATE_STATE, ATTACH_EVIDENCE, ARCHIVE_NODE), a typed payload, confidence score, and observability label. This operational representation is designed for streaming execution: operations are applied in order to a persistent graph store.

Step 4: Streaming Merge and State Management. The memory manager applies operations chronologically to a NetworkX MultiDiGraph. New nodes are created; duplicate or similar nodes are merged (based on label and node type overlap); intent states are updated as new evidence arrives (e.g., "plan to meet Sam" transitions from planned to ongoing when "on my way to meet Sam" is observed); and old or superseded information is archived rather than deleted, preserving provenance.

The key insight behind MCoC is that this decomposition serves three purposes simultaneously: it prevents over-extraction by requiring explicit candidate generation and typing before graph insertion; it makes uncertainty explicit through per-operation confidence and observability; and it enables streaming consistency by representing memory updates as an append-only log of operations that can be replayed, inspected, and rolled back.

4.5 AMR: Adaptive Memory Reasoning

AMR is designed on the principle that the persistent memory graph (built during memorization) and the query-time reasoning structure (built during control) should be architecturally separate. The persistent graph captures what the system believes to be true about the user's intents over time; the reasoning scaffold captures what the system needs to figure out to answer a specific query.

Step 1: Query Routing. The incoming natural language query is classified into one of six types: status_query (current state of intents), temporal_multi_hop (when something occurs relative to other events), update_conflict (changes, cancellations, contradictions), next_intent_prediction (what the user is likely to do next), short_horizon_planning (suggesting a schedule or sequence of actions), or acoustic_disambiguation (queries about sounds or environmental context). The router uses keyword matching in rule-based mode or LLM classification in local_hf mode.

Step 2: Scaffold Construction. Based on the query type, a small subgoal DAG is constructed. For example, a status query generates subgoals: (g1) identify relevant intents from the graph; (g2) collect supporting evidence for each intent; (g4) self-check completeness. A temporal query adds (g3) resolve temporal ordering and constraints. Each subgoal has defined dependencies, forming a topological execution order.

Step 3: Dependency-Aware Retrieval. Subgoals are executed in topological order. At each step, the retrieval engine searches the persistent memory graph for relevant nodes (matching query tokens against node labels and attributes), retrieves connected neighborhoods (following edges from matched nodes), and fetches supporting evidence (transcript snippets and acoustic tags linked via grounded_by edges). This is fundamentally different from naive GraphRAG: the retrieval target at each step depends on what was discovered in previous steps.

Step 4: Rolling Memory and Self-Check. Retrieved evidence is accumulated in a compact rolling summary to prevent context explosion. Before generating the final answer, a self-check step verifies that

all required subgoals have been resolved. If evidence is insufficient, the system can either re-retrieve with expanded search, answer with explicit uncertainty, or abstain completely—with the abstention decision recorded in the output for evaluation.

Output Format. Every reasoning result includes the answer text, the query type, the reasoning scaffold used, the graph trace (nodes and edges retrieved), the evidence trace (transcript snippets and acoustic tags used), a confidence score, and an abstained flag. This traceability is essential for research: it enables fine-grained evaluation of which retrieval steps succeeded or failed.

5. Implementation

5.1 Codebase Organization

The LongAI implementation follows a modular Python package structure designed for research reproducibility and extensibility. The codebase is organized into 12 subpackages under the `longai/` namespace, with CLI entry points in `scripts/` and YAML configuration files in `configs/`:

LongAI/	
└ longai/	
└ schema/models.py	# Pydantic models: EvidencePack, MemoryNode, MemoryEdge, GraphWriteOperation, ReasoningResult
└ data/	# Subset selection (ffprobe scanning), audio extraction (ffmpeg), session manifest building
└ tools/	# Perception wrappers with dual mock/local_hf backends
└ └ base.py	# ToolConfig dataclass + BaseTool ABC
└ └ vad.py	# Silero VAD (local_hf) / energy threshold (mock)
└ └ asr.py	# faster-whisper (local_hf) / duration mock (mock)
└ └ speaker_role.py	# Keyword heuristic + LLM extension point
└ └ scene.py	# AST AudioSet classifier (local_hf) / energy-ZCR (mock)
└ └ events.py	# AST AudioSet classifier (local_hf) / energy mock (mock)
└ construction/	# MCoC pipeline
└ └ evidence_pack.py	# Fuses tool outputs into EvidencePack objects
└ └ candidates.py	# Keyword (mock) / LLM (local_hf) candidate spotting
└ └ observability.py	# Rule (mock) / LLM (local_hf) observability typing
└ └ ops.py	# GraphWriteOperation factory
└ └ mcoc.py	# Orchestrator: EvidencePack → write operations
└ memory/	# Graph store + streaming manager
└ └ graph_store.py	# NetworkX MultiDiGraph with operation log
└ └ manager.py	# EvidencePacks → memory graph builder
└ reasoning/	# AMR pipeline
└ └ router.py	# Query type classification
└ └ scaffold.py	# Subgoal DAG builder
└ └ retrieval.py	# Token-match (mock) / LLM-guided retrieval
└ └ amr.py	# Orchestrator: query → ReasoningResult
└ eval/	# Metric implementations (tool, memory, reasoning)
└ train/	# Writer SFT (TRL LoRA), reasoner SFT/RL scaffolds
└ utils/	# I/O (JSON/JSONL), YAML config, LLM wrapper, hashing
└ configs/	# YAML: dataset, pipeline, training, evaluation
└ scripts/	# 17 CLI scripts for each pipeline stage
└ data/	# Pilot manifests, annotations, audio
└ artifacts/	# Pipeline outputs (perception, memory, reasoning, eval)
└ tests/	# Unit tests for schema, memory, reasoning

5.2 Data Preparation Pipeline

We selected a pilot subset of 5 sessions from the Ego4D video dataset [14]. The full dataset contains 9,104 MP4 files in the `video_540ss` split. The selection script (`select_pilot_subset.py`) uses `ffprobe` to scan candidate videos, filtering for those with audio tracks and durations between 300 and 3,600 seconds. From eligible candidates, 5 sessions are selected sequentially. The extraction script (`extract_audio.py`) invokes `FFmpeg` to produce 16kHz mono WAV files. The session manifest builder (`build_session_manifests.py`) records source paths, WAV paths, durations, and wearer identifiers.

Table 1: Pilot subset characteristics (5 sessions, 2.7 hours total audio).

Session	Source Video	Duration (s)	WAV (MB)	VAD Segments
session_0000	000cd456...mp4	958	30	34

session_0001	002ad105...mp4	1,063	33	126
session_0002	002c3b5c...mp4	2,758	85	2
session_0003	002d2729...mp4	3,571	109	1
session_0004	0049fdd8...mp4	1,363	42	117
Total	—	9,713 (2.7h)	297	280

The sessions exhibit substantial diversity in speech content: sessions 0000 and 0004 contain conversational speech with dozens of segments, while sessions 0002 and 0003 are nearly silent (2 and 1 segments respectively in 46 and 60 minutes of audio), demonstrating the speech sparsity characteristic of egocentric recordings.

5.3 Audio Perception Tool Integration

Each perception tool implements a common interface (`BaseTool`) with configurable backend (`local_hf` for real models, `mock` for rule-based stubs) and deterministic JSON caching. This design enables fast iteration: switching between mock and real backends requires only a flag change, and cached results prevent recomputation when re-running downstream stages.

VAD (Voice Activity Detection). The `local_hf` backend uses Silero VAD v6.2.1, a pre-trained ONNX model with a 16kHz sampling rate requirement. The model achieves real-time processing on CPU, outputting speech segments with per-segment confidence scores. On session_0000 (958s audio), Silero VAD found 34 segments totaling 18.3 seconds of speech. The mock backend uses an energy-threshold heuristic for debugging. All 280 detected segments across 5 sessions sum to 703.8 seconds of speech (7.2% of total audio).

ASR (Automatic Speech Recognition). The `local_hf` backend uses faster-whisper v1.2.1, which wraps OpenAI's Whisper model in the CTranslate2 inference engine for efficient GPU inference. We evaluated two model sizes: `tiny` (39M parameters, ~15s for 16min audio on A6000) and `base` (74M parameters, ~25s). The base model provides word-level timestamps, which are aligned to VAD segment boundaries to produce per-segment transcriptions. ASR confidence varies significantly across sessions: session_0004 achieves 63% nonempty segments with 0.486 average confidence, while session_0001 has only 10% nonempty, indicating that many detected VAD segments contain unintelligible or extremely short speech.

Speaker Role Attribution. The current implementation uses an improved keyword heuristic based on ASR text: first-person pronouns and possessive constructions (I, I'm, I'll, my, we, me) indicate `WEARER` speech; second-person constructions (you, you should) and reported speech markers (he said, she said) indicate `OTHER`; remaining segments are `UNKNOWN`, with every 9th unknown segment labeled `MIXED` for diversity. The tool interface accepts an optional WAV path for future acoustic diarization integration (pyannote.audio or Qwen2.5-Omni).

Scene and Event Classification. Both use the MIT Audio Spectrogram Transformer (`MIT/ast-finetuned-audioset-10-10-0.4593`), a vision transformer adapted to audio spectrograms and fine-tuned on AudioSet's 527-class ontology. For scene classification, AudioSet predictions are mapped to 8 LongAI categories (office, meeting_room, home, kitchen, street_outdoor, vehicle, cafe_shop, other) via keyword matching against AudioSet label descriptions. For event detection, predictions are mapped to 8 event types (typing, door, footsteps, traffic, kitchen_activity, appliance, crowd_chatter, silence_quiet) with a 0.15 confidence threshold. The model processes context windows extending 5 seconds on each side of every speech segment.

5.4 MCoC Implementation

The current MCoC pipeline uses keyword-based candidate spotting with a dictionary of 6 intent-triggering patterns (meeting → "attend meeting," call → "make call," schedule → "check schedule," prepare → "prepare materials," finish → "finish pending task," buy → "buy item"). Observability classification is rule-based: explicit markers ("I will," "need to," "must," "going to") indicate `EXTRACTED`; ASR confidence above 0.7 indicates `INFERRED`; otherwise `AMBIGUOUS`. A fully implemented LLM-based backend using Qwen2.5-7B-Instruct is available via the `--backend local_hf` flag, providing intent extraction, observability typing, and structured operation generation through few-shot prompting with Pydantic schema constraints.

The memory graph store uses NetworkX's MultiDiGraph, supporting parallel edges with different relation types. Each `GraphWriteOperation` is applied atomically, with an append-only history log recording every operation for reproducibility. The store supports node merging (combining duplicate intents across segments), state updates (transitioning intent states as new evidence arrives), and archival (marking superseded nodes rather than deleting them).

5.5 AMR Implementation

The AMR module implements both rule-based and LLM-based backends. The rule-based router uses keyword matching for query type classification; the scaffold builder constructs a 2-4 node DAG per query type; retrieval performs token overlap matching between query terms and graph node labels/attributes; and the answer generator produces template-based responses listing relevant intents and episodes. The LLM backend (Qwen2.5-7B-Instruct) uses the full AMR flow with structured prompting: query routing returns one of 6 types; scaffold construction generates a JSON subgoal structure; retrieval is LLM-guided with the model selecting which graph neighborhoods to explore; and answer generation includes explicit confidence estimation and abstention decisions.

5.6 Hardware and Environment

All experiments were conducted on a server with 8x NVIDIA RTX A6000 GPUs (48 GB VRAM each), running PyTorch 2.5.1 with CUDA 12.1 in the `longai` conda environment. GPU 3 was primarily used for inference. The environment includes: transformers 4.57.6, faster-whisper 1.2.1, silero-vad 6.2.1, torch 2.5.1, networkx 3.2, and pydantic 2.7. FFmpeg 4.2.7 is used for audio extraction.

6. Evaluation

6.1 Perception Pipeline Statistics

Table 2: Per-session ASR quality and coverage.

Session	Total Segments	Nonempty ASR	Nonempty Ratio	Avg ASR Confidence
session_0000	34	29	0.853	0.623
session_0001	126	13	0.103	0.283
session_0002	2	0	0.0	0.3
session_0003	1	0	0.0	0.3
session_0004	117	74	0.632	0.486

ASR coverage varies dramatically: session_0004 achieves 63.2% nonempty segments, while session_0001 has only 10.3% nonempty despite having the most segments (126). Manual inspection reveals that session_0001's segments are predominantly very short (<0.5s) and contain fragments that the base model cannot reliably transcribe. Session_0004's segments average 2-5 seconds, providing sufficient context for transcription. This highlights a critical pipeline dependency: VAD segment duration directly impacts downstream ASR quality.

Table 3: Speaker role distribution per session.

Session	Role Distribution (counts)
session_0000	{'wearer': 7, 'unknown': 24, 'mixed': 3}
session_0001	{'mixed': 14, 'unknown': 112}
session_0002	{'mixed': 1, 'unknown': 1}
session_0003	{'mixed': 1}
session_0004	{'mixed': 10, 'unknown': 74, 'wearer': 25, 'other': 8}

Session_0004 shows the most informative speaker role distribution: 25 wearer segments (21%), 8 other-person segments (7%), 10 mixed segments, and 74 unknown. The high unknown rate reflects the difficulty of text-only speaker classification—many conversational utterances lack clear first/second-person markers

that the keyword heuristic can detect. Acoustic speaker diarization would substantially improve this classification.

Table 4: Scene classification distribution.

Session	Scene Distribution
session_0000	{'meeting_room': 34}
session_0001	{'meeting_room': 126}
session_0002	{'meeting_room': 2}
session_0003	{'other': 1}
session_0004	{'meeting_room': 117}

The AST classifier labels 99.3% of segments as "meeting_room." This reflects a genuine property of the pilot data—the detected speech segments are embedded in conversational contexts—but also reveals a limitation of the generic AudioSet mapping: many egocentric indoor environments (living rooms, kitchens with conversation) are not well-represented in the 8-class LongAI ontology. Domain-specific scene classifiers trained on egocentric audio would provide finer-grained discrimination.

6.2 Memory Graph Construction Results

Table 5: Memory graph statistics across sessions.

Session	Total Nodes	Total Edges	Intent Nodes	Episode Nodes	Entity Nodes
session_0000	35	25	1	34	0
session_0001	127	10	1	126	0
session_0002	2	0	0	2	0
session_0003	1	0	0	1	0
session_0004	120	40	3	117	0

Graph size correlates strongly with VAD segment count. Session_0004 produced the richest graph (120 nodes, 40 edges) with 3 detected intents: "general ongoing intent" (default intent assigned to segments without keyword matches but with sufficient text length), "make call" (triggered by the keyword "call"), and "buy item" (triggered by "buy"). Sessions 0002 and 0003, each with minimal speech, produced near-empty graphs. The keyword-based approach achieves high precision for its limited vocabulary—the "buy item" intent correctly reflects the shopping list conversation in session_0004—but misses numerous intents present in the conversation, such as visiting a restaurant, meeting "Bruce," and examining a cocktail menu.

The entity node count is zero across all sessions because the keyword-based candidate spotter only detects entities when they match the small set of token patterns ("meeting," "teammate," "home," "office"), and the ASR output from these sessions does not contain these exact tokens. This represents a significant gap in the current implementation that LLM-based extraction is designed to address.

6.3 Reasoning Results

Table 6: Per-session reasoning statistics (3 standard queries each).

Session	Queries	Avg Confidence	Abstain Rate	Avg Graph Trace Length
session_0000	3	0.62	0.0	4.7
session_0001	3	0.62	0.0	4.7
session_0002	3	0.62	0.0	3.0
session_0003	3	0.62	0.0	2.0
session_0004	3	0.62	0.0	6.0

Three standard queries were evaluated per session: (Q1) "What is the current status of user intent?" (status_query), (Q2) "What is the next intent?" (next_intent_prediction), and (Q3) "What changed or was

canceled?" (update_conflict). The rule-based AMR reports uniform 0.62 confidence with 0% abstention across all sessions, reflecting the baseline's inability to calibrate confidence or recognize when evidence is insufficient. Sessions with content (0000, 0001, 0004) return graph traces of 4.7-6.0 items on average; sessions with minimal content (0002, 0003) return shorter traces but do not abstain.

For session_0004, the status query (Q1) correctly retrieved all three detected intents. However, the next-intent prediction query (Q2) returned episode nodes rather than predicting future intents, because the token-match retrieval matches "next" and "intent" to episode labels rather than performing actual temporal reasoning. The update/conflict query (Q3) returned general graph nodes without identifying specific state changes. These limitations stem from the rule-based retrieval's reliance on surface-form token matching rather than semantic understanding of intent state transitions.

6.4 Qualitative Case Study: Session 0004

Session 0004 provides the richest content for qualitative analysis. The 23-minute recording captures a conversation between the wearer and an interlocutor discussing a shopping list. The following excerpts illustrate the diversity of intent-relevant content spread across the session:

[17s-21s] speaker=unknown scene=meeting_room

"this is definitely somebody else's shopping list."

[27s-31s] speaker=wearer scene=meeting_room

"is that? I noticed they listed it separately as if it's not a beer. Is it"

[32s-35s] speaker=unknown scene=meeting_room

"type of beer or something? Or they're just calling it a beer?"

[36s-40s] speaker=unknown scene=meeting_room

"That's different. Bruce, what is that? That sounds like Chicago."

[63s-65s] speaker=wearer scene=meeting_room

"I guess they don't want you to buy the alcohol."

[70s-73s] speaker=other scene=meeting_room

"that's one way to stop you from doing something bad."

[1071s-1076s] speaker=wearer scene=meeting_room

"was just my mental marker. It's on the same street as that."

[1076s-1080s] speaker=mixed scene=meeting_room

"But it wasn't. This one isn't Scottish. This is the one that had the nice hot toddy."

[1082s-1084s] speaker=unknown scene=meeting_room

"Cocktails and a southern creole menu."

This conversation reveals multiple personal intents distributed across 1,084 seconds: (1) **shopping task**—the wearer is examining a shopping list that includes beer and other items; (2) **purchase constraint**—the interlocutor notes restrictions on alcohol purchases; (3) **restaurant visit**—discussion of "the one that had the nice hot toddy" and "cocktails and a southern creole menu" suggests a planned or remembered restaurant outing; (4) **social meeting**—the mention of "Bruce" indicates interaction with a specific person. The keyword-based MCoC captures only the generic "buy item" intent from this rich conversation, underscoring the need for LLM-based extraction.

6.5 Ablation: ASR Model Size Comparison

Table 7: ASR model size ablation on session_0004.

Model	Parameters	Nonempty Ratio	Avg Confidence	Inference Time
faster-whisper tiny	39M	58.1%	0.412	~15s
faster-whisper base	74M	63.2%	0.486	~25s

The base model provides a 5.1 percentage point improvement in nonempty segment ratio and 0.074 higher average confidence compared to the tiny model, at a 67% increase in inference time (15s to 25s for 23 minutes of audio). For the target deployment scenario (a few A6000 GPUs), this tradeoff strongly favors the base or even small model, as ASR quality improvements cascade through all downstream stages. The small model (244M parameters) is expected to provide further quality gains and is recommended for the next iteration.

6.6 Answer to Research Questions

RQ1 (Perception Feasibility): YES. The integrated perception pipeline successfully processes all 5 Ego4D sessions through VAD (Silerio), ASR (faster-whisper), speaker role attribution (keyword heuristic), scene classification (AST), and event detection (AST), producing 280 timestamped EvidencePacks. The pipeline is fully cached and resumable.

RQ2 (Memory Construction Quality): PARTIAL. MCoC produces structured memory graphs from real evidence packs (120 nodes and 40 edges for the richest session). However, the keyword-based candidate spotter achieves high precision but low recall—capturing only 3 intents from a conversation containing at least 4-5 distinct intent-bearing topics. LLM-based extraction is the clear next step for improving recall.

RQ3 (Reasoning Effectiveness): LIMITED. AMR answers queries with relevant graph traversal results, but the rule-based implementation produces templated responses with uniform confidence and no abstention capability. The LLM-based backend (implemented but not fully evaluated due to GPU constraints) is designed to address these limitations.

RQ4 (Tool Contributions): VAD quality (segment duration and count) is the primary determinant of downstream graph richness. ASR model size provides a measurable boost in transcription coverage. Scene and event classifiers, using generic AudioSet models, provide limited discriminative power for egocentric audio. Speaker role from text alone has high unknown rates.

RQ5 (Practical Deployment): The modular backend design (mock/local_hf switching), deterministic caching, and configurable model sizes enable effective experimentation on limited GPU resources. The 5-session pilot processes in under 5 minutes on a single A6000. Scaling to 100+ sessions would require batch processing and GPU scheduling but is architecturally supported.

7. Discussion

7.1 Key Findings and Implications

This project demonstrates that audio-first personal intent memory construction from real egocentric data is feasible but faces several fundamental challenges. The most critical finding is that **speech sparsity is the primary bottleneck**: only 7.2% of our pilot audio contains detectable speech, and within that, ASR quality and segment duration heavily influence downstream memory quality. This has implications for wearable AI design—systems must be architected to handle long stretches of silence or non-verbal audio without degradation, and memory updates should be triggered by speech events rather than continuous processing.

The keyword-based MCoC serves as a useful baseline but fundamentally cannot capture the diversity of natural language expressions of intent. Human intent is expressed through a wide variety of linguistic constructions (conditional statements, indirect requests, implicit commitments) that resist simple keyword matching. The LLM-based backend addresses this, but at significant computational cost: loading a 7B parameter model for per-segment inference would require batched processing and careful GPU memory management.

Scene and event classification from generic AudioSet models provides coarse environmental context but lacks the specificity needed for egocentric intent disambiguation. Future work should explore domain-adapted classifiers or the use of multimodal audio-text models like Qwen2.5-Omni that can jointly reason about speech content and acoustic context.

7.2 Relationship to M3-Agent and LogicRAG

LongAI's memorization/control decomposition directly inherits from M3-Agent [3], but the adaptation to audio-first, speech-centric processing represents a non-trivial architectural shift. M3-Agent's fixed clip-based processing would waste ~93% of computation on non-speech audio in our setting. The speech-

centric design is more efficient but creates an uneven temporal structure that complicates streaming memory updates—intent evidence arrives in bursts separated by long gaps, requiring the memory manager to maintain state across variable-duration intervals.

The AMR framework's use of query-time reasoning scaffolds is inspired by LogicRAG [13], but applied to a different substrate: LogicRAG operates over logical expressions derived from text, while AMR operates over a persistent intent graph with linked multimodal evidence. The key adaptation is the integration of evidence-level retrieval (transcript snippets, acoustic tags) alongside graph-level retrieval (nodes and edges), enabling the reasoner to "drill down" from abstract intent nodes to the raw perceptual data that supports them.

7.3 Limitations and Future Work

This work has several important limitations that define the trajectory for future development:

Dataset Scale. The pilot evaluation uses only 5 sessions (2.7 hours). Statistical conclusions require scaling to at least 100 sessions across diverse activities, environments, and wearers. The Ego4D dataset provides sufficient scale for this expansion.

Ground Truth Absence. No human-annotated ground truth exists for tool outputs, memory graphs, or reasoning results. All current evaluation is system-internal (coverage, confidence, graph size). The next phase requires tool-level annotation (VAD boundaries, ASR transcripts, scene labels) and graph-level annotation (intent nodes, edges, state transitions) for at least a subset of sessions. Annotation templates and schemas are already implemented in the codebase.

Keyword MCoC Limitations. The 6-keyword dictionary captures a tiny fraction of intent expressions. LLM-based extraction using Qwen2.5-7B-Instruct is implemented but was not fully evaluated due to GPU memory constraints on the shared server. Supervised fine-tuning of a smaller model (e.g., 1.5B parameters) on graph-write trajectories could provide a practical middle ground.

Speaker Diarization Gap. Text-only speaker role classification has a 63% unknown rate. Integration of pyannote.audio or Qwen2.5-Omni for acoustic speaker attribution would substantially improve memory quality by correctly attributing intent statements to the wearer.

Reasoning Evaluation. Without ground-truth QA pairs, reasoning evaluation is limited to manual inspection. Creating a benchmark of 50-100 annotated QA pairs with expected answers, evidence references, and acceptable abstention conditions is essential for rigorous reasoning evaluation.

Privacy Considerations. Continuous egocentric audio recording raises significant privacy concerns. Any practical deployment would require on-device processing, local storage only, and user-controlled data retention policies. These constraints are compatible with the modular tool architecture (models like Silero VAD and Whisper-tiny can run on-device) but require explicit design attention.

8. Conclusion

We have presented LongAI, an audio-first research system for streaming personal intent memory construction from long egocentric audio. The system implements a three-stage architecture: (1) audio perception using five specialized tools (VAD, ASR, speaker role, scene, events) with dual mock/real backends; (2) streaming memory construction via Multimodal Chain-of-Construction (MCoC), which incrementally builds a structured intent memory graph with explicit confidence and observability tracking; and (3) Adaptive Memory Reasoning (AMR), which performs query-time reasoning using dependency-aware retrieval over both graph and evidence levels.

The system was evaluated on a pilot subset of 5 real Ego4D sessions (2.7 hours, 280 detected speech segments) using silero-vad, faster-whisper-base, and AST-based audio classifiers on NVIDIA A6000 GPUs. The pipeline successfully extracts evidence packs, constructs queryable memory graphs (up to 120 nodes and 40 edges per session), and answers intent-related queries. Qualitative analysis of the richest session revealed multiple personal intents (shopping, restaurant visit, social meeting) distributed across a 23-minute conversation, with the keyword-based MCoC capturing the "buy item" intent but missing others—motivating LLM-based extraction for future work.

This project establishes the foundational infrastructure for a publishable research system addressing the problem of streaming personal intent memory from egocentric audio. The modular, configuration-driven codebase supports rapid experimentation, and the dual backend architecture enables smooth transitions

from prototyping with mocks to research with real models. The complete source code, including all pipeline scripts, evaluation tools, and this report, is available at the project repository.

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